Queen Mary
University of London

EBU5376
Digital Signal Processing

Week 1 – Tutorial 2

REMINDER: Nyquist-Shannon Theorem

- Theorem: To ensure perfect reconstruction, a signal must be sampled at a rate **greater than twice the highest frequency** in the signal.

- If the highest frequency component is B Hz, then a sufficient sampling rate is $f_s > 2B$
- The threshold $f_s = 2B$ is the Nyquist rate
- In other words, for a fixed sampling rate f_s, this is enough for any signal with a maximum frequency component $B < \frac{f_s}{2}$.
- The frequency $B = \frac{f_s}{2}$ is the Nyquist frequency

Reminder Aliasing

- Given an under-sampled signal, it is aliased to the following frequencies

$$f_a = |nf_s - f| \ , \qquad n = 0, \pm 1, \pm 2, \ldots$$

- In the equation, f_s is the sampling rate, f is the (fundamental) signal frequency
- Usually, we are only interested in the lowest positive frequency that the signal is aliased to
  - The lowest frequency is the one that can be sampled without aliasing

Question

We generally assume that audio signals (that is, ones that humans can hear) have frequencies in the range 20Hz to 20kHz (although many people will not be able to hear a 20kHz tone……but that is another story).

What does the Nyquist sampling theorem state is the minimum sampling theory for an audio signal?

Question

Suppose we sample an audio signal at 44.1 kHz (this is usual for music signal processing).

If we do not bandlimit the audio signal before sampling, find the 5 lowest frequencies that are aliased to 15 kHz. That is, find 5 higher frequencies whose samples will appear to be 15 kHz.

Question

Suppose we sample an audio signal at 44.1 kHz.

Find the alias frequency of a noise frequency of 100 kHz.

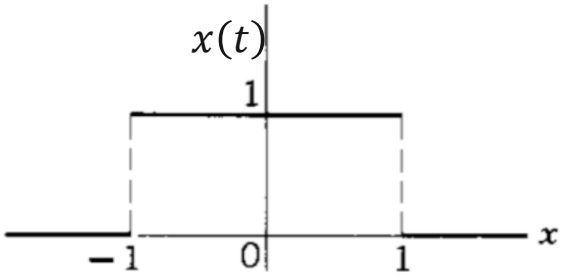
Fourier Transform: Reminder

- The Fourier Transform $X(\omega)$ is a complex-valued function.

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_{-\infty}^{\infty} x(t)\,(\cos\omega t - j\sin\omega t)\,dt$$

  - The real part $Re\big(X(\omega)\big)$ represents the contribution of the **cosine waveforms** at frequency ω
  - The imaginary part $Im\big(X(\omega)\big)$ represents the contribution of the **sine waveforms** at frequency ω
  - The magnitude $|X(\omega)|$ indicates the overall amplitude of the frequency component at ω
  - The phase $\angle X(\omega) = \tan^{-1}\left(\frac{Im\big(X(\omega)\big)}{Re\big(X(\omega)\big)}\right)$ describes the relative shift of the frequency component

Fourier Transform Example Reminder



- <u>Example</u>: $x(t) = \begin{cases} 1, & -1 < t < 1 \\ 0, & |t| > 1 \end{cases}$

- <u>Solutions</u>: $\mathcal{F}\big(x(t)\big) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-1}^{1} e^{-j\omega t} dt$

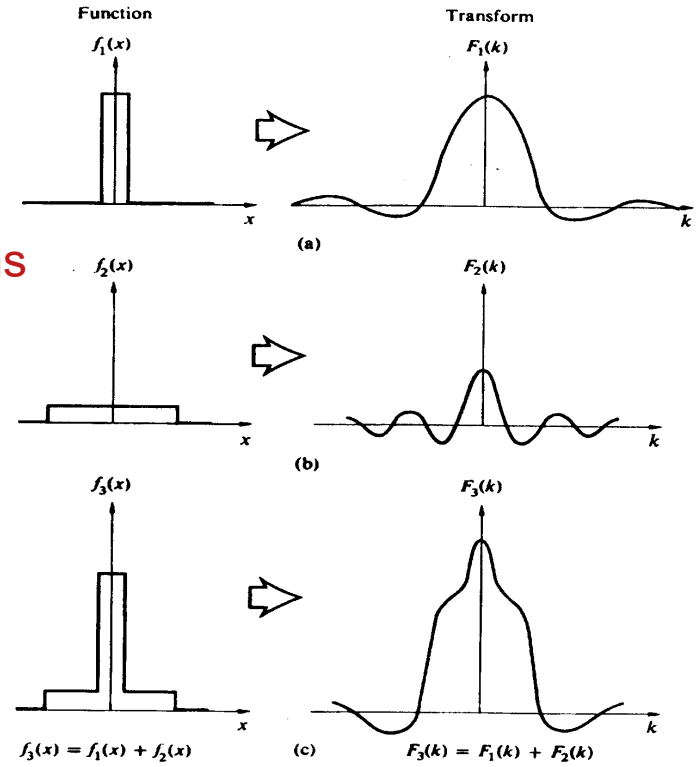
$$= -\frac{1}{j\omega} e^{-j\omega t} \Big|_{-1}^{1} = \frac{e^{j\omega} - e^{-j\omega}}{j\omega} = \frac{\cos\omega + \mathrm{j}\sin\omega - (\cos\omega - \mathrm{j}\sin\omega)}{j\omega}$$

$$= \frac{2j\sin\omega}{j\omega} = \frac{2\sin\omega}{\omega} = 2sinc(\omega)$$

Properties of Fourier Transform

- The Fourier Transform exhibits many useful properties.
- For example:
  1. **Linearity**
  2. Time-shift
  3. Frequency-shift
  4. Time scaling
  5. Time differentiation
  6. Time integration
  7. Convolution
- Formulations and proofs of these properties can be found in Appendix C

The Fourier Transform of a sum of two functions is the sum of the Fourier Transforms of each function



Properties of Fourier Transform Reminder

- The Fourier Transform exhibits many useful properties.
- For example:
  1. Linearity
  2. Time-shift
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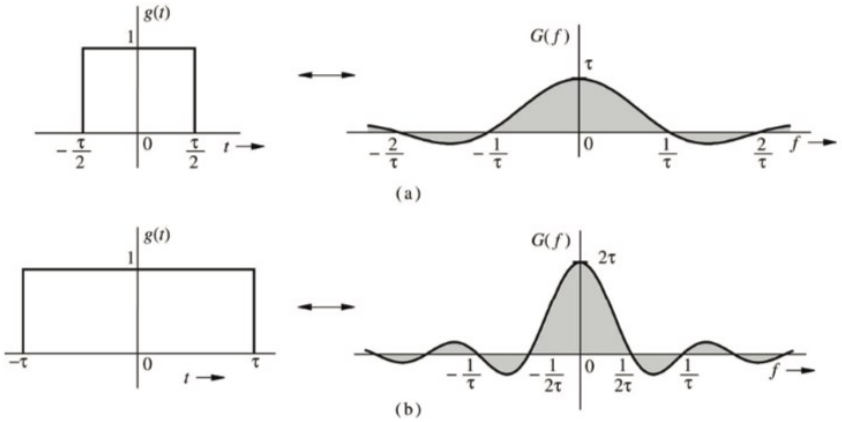
Reducing the time-domain duration of a waveform causes its spectrum to spread out in the frequency domain, dictating a larger system bandwidth.



Fourier Transform Time scaling example

$$x(t) = \begin{cases} 1, & -1 < t < 1 \\ 0, & |t| > 1 \end{cases}$$

$$\mathcal{F}\big(x(t)\big) = 2sinc(\omega)$$

This is known as rectangle function *rect(t)*

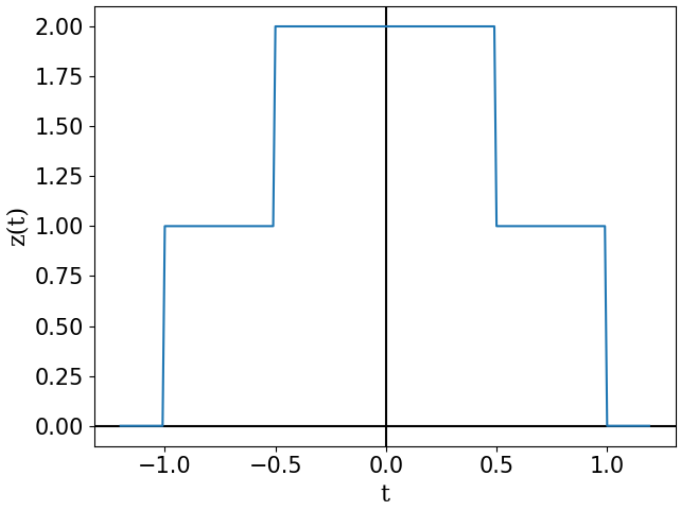
$$y(t) = \begin{cases} 1, & -1/2 < t < 1/2 \\ 0, & |t| > 1/2 \end{cases}$$

$$\mathcal{F}\big(y(t)\big) = ?$$

If you are clever you answer this without calculating integrals

Fourier Transform

$$z(t) = \begin{cases} 2 & |t| < \frac{1}{2} \\ 1 & \frac{1}{2} \leq |t| < 1 \\ 0 & |t| \geq 1 \end{cases}$$



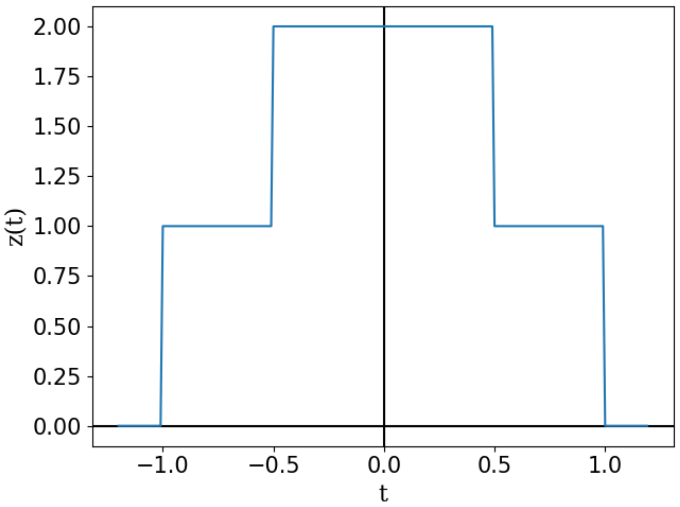
$$\mathcal{F}\big(x(t)\big) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-1}^{-1/2} e^{-j\omega t} dt + \int_{-1/2}^{1/2} e^{-j\omega t} dt + \int_{1/2}^{1} e^{-j\omega t} dt$$

Will work but sooooo annoying and fiddly

Fourier Transform

$$z(t) = \begin{cases} 2 & |t| < \frac{1}{2} \\ 1 & \frac{1}{2} \leq |t| < 1 \\ 0 & |t| \geq 1 \end{cases}$$



$$x(t) = \begin{cases} 1, & -1 < t < 1 \\ 0, & |t| > 1 \end{cases}$$

$$y(t) = \begin{cases} 1, & -1/2 < t < 1/2 \\ 0, & |t| > 1/2 \end{cases}$$

$$z(t) = x(t) + y(t)$$

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